

Article

Hydrogen-Rich Syngas Production Based on a Co-Gasification Process Coupled to a Water–Gas Shift Reactor Without Steam Injection

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Abstract: Future decarbonized applications that rely on renewable and carbon-dioxide-neutral hydrogen production could benefit from the gasification of waste to produce hydrogen. In the current study, an Aspen Plus[®] model was developed by coupling a co-gasification model to a water–gas shift (WGS) model. The feedstock employed in the simulations was a blend of municipal solid waste (MSW) and biomass from Morocco. A parametric assessment was conducted to analyze the effect of the steam-to-feedstock ratio (SFR) on the syngas composition and the WGS reactor temperature. This study also presents a comparison between the results of the gasification process before and after the WGS reactor, using air and steam as the gasifying agent. The results show an increase in hydrogen volumetric percentage for higher steam-to-feedstock ratios in the gasifier. Moreover, the inclusion of a WGS reactor enhances hydrogen and carbon dioxide while reducing the amount of carbon monoxide in the syngas for both air and steam as the gasifying agents. It can be concluded that a co-gasification process can be intensified by coupling it to a WGS reactor without steam injection to produce hydrogen-rich syngas with reduced operational expenditures.

Keywords: co-gasification; hydrogen rich syngas; water–gas shift reactor; Aspen Plus



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1. Introduction

The optimal use of biomass for energy production requires good knowledge and mastery of the raw material in terms of its technical characteristics and potential. According to the National Roadmap for Energy Valorization from Biomass, an assessment of the sectors of agriculture, forestry, municipal solid waste (MSW), and wastewater has identified the main energy potential from biomass in Morocco [1]. The total technical primary energy potential in 2015 was roughly 13.4 TWh per year, of which 6.6 TWh/year came from agriculture, 3.5 TWh/year came from forestry, 3.1 TWh/year came from waste, and 0.2 TWh/year came from wastewater [1]. These statistics show the importance of the agricultural and waste sectors for their technical energy potential, as well as the concentration of potential in the northern regions of Morocco. The regions of Fes-Meknes and Casablanca-Settat have the greatest potential. While the potential in Casablanca-Settat is based mainly on organic household waste and green waste, the potential in Fes-Meknes is characterized by potential from agriculture [2]. Morocco has approximately 7 million head of cattle and a forestry area of approximately 9 million hectares. This represents a large biomass potential, which, along with the MSW, can be used as feedstock in biomass power plants [3]. Population growth, rising living standards, and urbanization all contribute to Morocco's rising waste production. MSW management is one of Morocco's most pressing environmental issues.

More than five million tons of MSW are generated each year, and a 3% yearly growth rate is forecasted for the country [4]. This means that Morocco's MSW production will rise to around 9.30 million tons in 2030 [2].

MSW disposal in Morocco can be solved by providing adequate resources and infrastructure. One of the suitable infrastructures to implement may involve the use of thermal technologies (e.g., incineration, pyrolysis, and gasification) to extract energy from MSW [5]. Biagini et al. [6] performed a comparative study to assess the suitability of combustion, gasification, electrolysis, and syngas separation processes for hydrogen generation from biomass. They concluded that a greater amount of hydrogen is generated when using the gasification/separation processes, followed by gasification/electrolysis, and lastly by combustion/electrolysis processes. Monteiro and Brito [7] reviewed hydrogen production technologies from fossil fuels and renewable energy sources. They concluded that the most economical technologies for hydrogen production are partial oxidation, steam methane reforming, coal gasification, and biomass gasification. Thus, the gasification of biomass has been considered as one of the most efficient pathways for hydrogen generation [8,9].

Waste gasification is a thermal process that occurs in a substoichiometric atmosphere at temperatures ranging from 800 to 1200 °C [10,11]. The gas produced by gasification is generally known as syngas, and it can be used to make chemicals or transformed into fuels like hydrocarbons or hydrogen [12]. Because of the complexity of the biomass gasification process, modeling and simulation are helpful tools as a first approach to a biomass gasification project [13]. In this regard, Aspen Plus® has been extensively used in the development of various gasification models to optimize hydrogen production, mainly because of its extensive library of properties and operating units [14].

Singh and Tirkey [15] investigated the steam plasma gasification of medical waste in Aspen Plus®. They concluded that if steam is used as the gasifying agent, it will increase the amount of hydrogen in the syngas. Bourguig et al. [16] enlarged the parametric analysis, developing a co-gasification plant model in Aspen Plus for hydrogen production from municipal solid waste and biomass mixtures. Their main findings show an increase in hydrogen molar fractions for higher temperatures and SFR ratios. Kakati et al. [17] developed experimental work and an Aspen Plus model of steam gasification to study the production of hydrogen-rich syngas. They obtained a maximum hydrogen yield of 37.12% for a steam-to-biomass ratio of 0.35. Cao et al. [18] enlarged the analysis to different gasifying agents (air, oxygen-enriched air, air/steam, and oxygen-enriched air/steam) when developing a detailed Aspen Plus® model for biomass gasification. They concluded that a rise in oxygen percentage leads to an increase in H₂ production. The maximum H₂ content was obtained when oxygen-enriched air/steam was used as a gasifying agent.

As can be seen, the optimization of hydrogen production in a gasification scenario is generally based on a parametric study, while process intensification is very rarely approached. The following are the sole examples of research on gasification process intensification for the increase of hydrogen production. Novais et al. [19] developed an air-blown gasification model in Aspen Plus®, integrating a water–gas shift reactor without catalysts to study its optimization. The gasifier was optimized for carbon monoxide or hydrogen production and compared to the steam gasification process. They concluded that coupling a WGS reactor with an air-blown biomass gasification process results in 52.5% hydrogen molar fractions in syngas while using lower steam flow rates than those of the steam gasification process. Pala et al. [20] used Aspen Plus® to create a simulation model of steam biomass gasification and subsequent syngas adjustment using a water–gas shift reactor. They concluded that the H₂ and CO concentrations were changed in such a way that the H₂/CO ratio was adjusted to a value close to the 2.15 needed for Fischer–Tropsch synthesis. Chu et al. [21] investigated hydrogen generation from syngas with a high carbon dioxide content using a water–gas shift reactor in Aspen Plus®. The Cu–Zn and Fe–Cr catalysts were used to increase hydrogen yield, and the influence of velocity, temperature, and H₂O/CO ratio were studied. They concluded that both catalysts are suitable for syngas water–gas shift reactions at high CO₂ concentrations. At 450 °C, the maximum value of H₂

increase occurs, which is unaffected by the increase in CO₂. Ersoz et al. [22] established an integrated model in Aspen Plus involving a bubbling fluidized bed reactor, a syngas cleaning and conditioning unit (hydrocarbon reforming and water–gas shift processes), and a pressure swing adsorption unit to isolate the hydrogen from the syngas stream. The syngas obtained had 38.8% H₂ and 1.65% CO. Mojaver and Khalilarya [23] developed an integrated system comprising a gasifier reactor, steam methane reformer, and water–gas shift reactor, intended to convert face mask waste into a hydrogen-rich syngas. Their results indicate a cumulative increase of 160% in hydrogen content and a reduction in methane content and carbon monoxide of 38.4% and 78.8%, respectively.

As previously noted, the current approach to enhancing the hydrogen content in the syngas involves the coupling of a WGS reactor with additional steam supply. Given that steam production is an energy-intensive and thus expensive process, the novelty of this work is the investigation into improving hydrogen production from a co-gasification process coupled to a water–gas shift reactor without steam injection, making use of the steam generated by moisture evaporation.

2. Materials and Methods

2.1. Feedstock Characterization

The feedstock selected to feed the co-gasification process is a blend of organic MSW and biomass characteristic of Moroccan waste products. The co-gasification feedstock was defined to maximize the hydrogen amount in the syngas according to the findings of Bourguig et al. [16]. Therefore, the feedstock mixture is comprised of 70% MSW and 30% biomass, the ultimate and proximate analyses of which are depicted in Table 1.

Table 1. Composition of the feedstock mixture [16].

Ultimate Analysis (wt.%, d.a.f.)		Proximate Analysis (wt.%)	
C	55.23	Volatile Matter	64.4
H	8.04	Fixed Carbon	12.6
N	1.99	Moisture	19.8
O	34.74	Ash	3.2

The oxygen content was calculated in Bourguig et al. [16] by subtracting it from the other elements in the ultimate analysis on a dry and ash-free basis, as shown in Table 1.

2.2. Model Description

The developed simulation model comprises two main processes: the co-gasification process and a water–gas shift process. The former is modeled according to the model developed in a previous publication by Bourguig et al. [16]. The latter is added in this study to the simulation model as a WGS reactor (WGSR). The entire model is built on thermodynamic equilibrium and on the following methods and assumptions [24,25]:

- The model is based on the non-stoichiometric method.
- The model is in a steady state.
- The processes are isothermal and isobaric.
- Char is assumed to contain only carbon.
- Gases behave as ideal gases.
- Ashes and tars are disregarded in the process.

Drying, pyrolysis, and gasification are the main steps used to simulate the co-gasification process. The Aspen Plus flowchart of the model developed by Bourguig et al. [16] is shown in Figure 1.

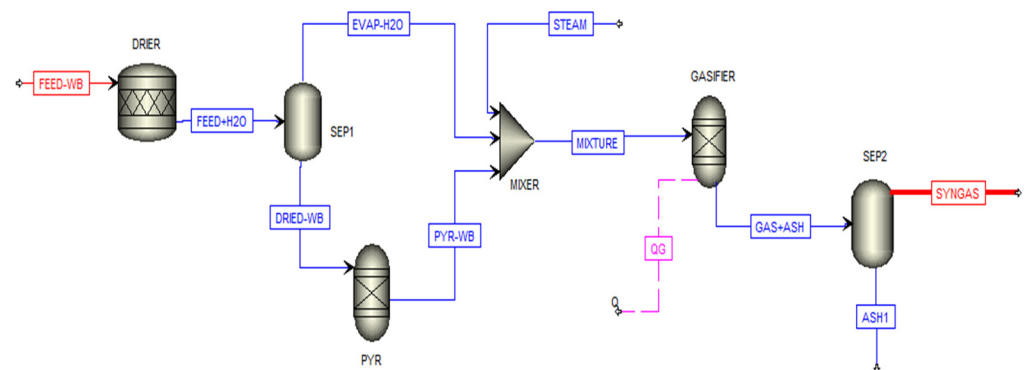


Figure 1. Flowchart of the co-gasification process developed in Aspen Plus [16].

The simulation employs two types of components: conventional and non-conventional. The thermodynamic properties of conventional components are computed using Peng–Robinson–Boston–Mathias’s equation of state (PR-BM). This model is suitable when working with gasification at high temperatures [26,27]. The density and enthalpy of non-conventional solid materials are computed, respectively, using the COALIGT and HCOAL-GEN models [28]. The wet MSW and biomass stream (FEED-WB) are fed into the dryer block, where they are identified as non-conventional components through ultimate and proximate analyses. The moisture in the feedstock is converted into the conventional component H₂O at the stoichiometric dryer block (DRER). The stream leaving the dryer block is supplied to the pyrolysis block (PYR) and modeled as a YIELD reactor. In this reactor, the feedstock is decomposed into volatiles, char, and ash. The feedstock is then supplied to the gasification block, comprising a Gibbs reactor (RGIBBS) along with steam, which is used as the gasifying agent in this process. A detailed description of this co-gasification model can be found in a previous publication by Bourguig et al. [16].

In this study, the co-gasification model is further developed by including a water–gas shift reactor (WGSR) to try to enhance the quality of the syngas in terms of hydrogen content, avoiding the addition of steam to the WGSR to decrease the operational expenditures. The new Aspen Plus flowchart of the entire model is shown in Figure 2.

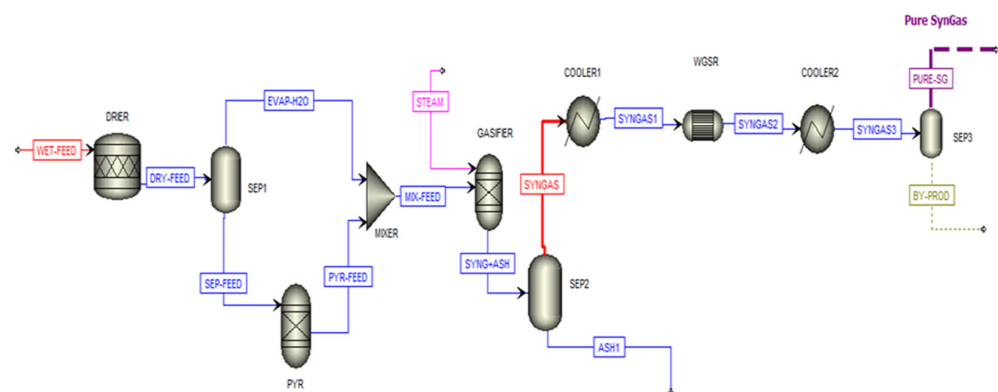
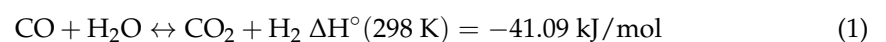


Figure 2. Flowchart of the co-gasification model coupled to a water–gas shift reactor.

The syngas coming from the separator (SEP2) enters the heat exchanger (COOLER1) to adjust its temperature to that of the WGSR. Then, the syngas is supplied to the reactor (WGSR). In this reactor, the water–gas shift reaction, also known as the Dussan reaction [29], occurs. It is a chemical reaction that converts a combination of carbon monoxide and steam into a combination of carbon dioxide and hydrogen, as follows [30]:



Equation (1) shows that the WGS reaction is reversible and moderately exothermic. To enable large-scale hydrogen production from syngas, a suitable catalyst must be used to promote the WGS reaction [30]. Pal et al. [31] offer a very complete overview of the various types of WGS reaction catalysts. Moreover, practical applications of the WGS reaction are conducted using two adiabatic stages: the high temperature shift followed by the low temperature shift [32]. Usually, iron-chromium and copper-zinc catalysts have been used to boost the reaction at high and low temperatures, respectively. Copper-based catalysts used for low temperature shift reactions are prone to poisoning by the sulfur compounds present in some biomass sources, such as the MSW used in this work, whereas the iron-based catalysts are more robust and sulfur-tolerant [33,34]. A detailed discussion on various high temperature and low temperature shifts and other metals and nanomaterial catalysts is offered by Pal et al. [31]. The WGSR is herein modeled based on a thermodynamic approach referred to as a Gibbs reactor (RGibbs), which models single-phase chemical equilibrium, or simultaneous phase and chemical equilibria (non-stoichiometric), by minimizing the Gibbs free energy [35]. The input parameters are the reactor temperature and pressure. After the WGS reaction, the produced gas goes through a second heat exchanger (COOLER2) to reject the surplus heat, which can be used, along with the heat rejected in the COOLER1, for district heating applications [36]. Finally, a separator (SEP3) is used to detach byproducts from pure syngas, which is rich in hydrogen. Table 2 shows the input parameters used in the WGS model.

Table 2. Aspen Plus® inputs used in the WGS model [37].

Unit Name	Block ID	Aspen Plus Unit	Input
Syngas cooling	COOLER	Heater	200 < T < 700 °C; 1 atm
WGS reactor	WGSR	RGibbs	200 < T < 700 °C; 1 atm
Separator	SEP	Flash	1 atm; Duty = 0 kW

2.3. Model Validation

The gasifier block model developed was validated against two experimental results of Jayah et al. [38] on rubber wood gasification under different conditions. Proximate and ultimate analyses of rubber wood are shown in Table 3.

Table 3. Proximate and ultimate analysis of rubber wood [38].

Proximate Analysis (wt.%, d.b.)		Ultimate Analysis (wt.%, d.b.)	
Volatile matter	80.1	C	50.6
Fixed carbon	19.2	H	6.5
Ash	0.7	O	42.0
		N	0.2

A comparison between the experimental results of Jayah et al. [38] for a temperature of 1100 K and the conditions is expressed in Table 4.

Table 4. Summary of gas analyses from three gasification experiments [32].

Experiment	Chip Size (cm)	Moisture (%)	Air/Fuel Ratio	H ₂	CO	CO ₂	N ₂
1	4.4	16.0	1.96	17.0	18.4	10.6	52.7
2	5.5	14.7	1.86	15.5	19.1	11.4	52.9

The deviation of the Aspen Plus results from the experimental results is quantified using the relative error, defined as follows:

$$\text{Rel. Error(\%)} = \frac{\text{Experimental value} - \text{Numerical value}}{\text{Experimental value}} \times 100 (\%) \quad (2)$$

The gasification model developed can be considered validated once it achieves results that agree well with the experimental data [38]. It is noted that the relative error is below 16.7% for the main gas species present in the syngas (Table 5). This can be considered a good approximation since the Aspen Plus model approach used in this work neglects some biomass gasification features, such as the particle size, chemical kinetics, and fluid dynamics. Earlier published works found similar relative errors [39,40].

Table 5. Comparison between Aspen Plus results and literature experimental results.

Data	H ₂	CO	CO ₂	N ₂
Experiment 1	17.0	18.4	10.6	52.7
Aspen Plus model (%)	15.78	20.64	9.72	53.86
Relative Error (%)	7.2	−12.2	8.3	−2.2
Experiment 2	15.5	19.1	11.4	52.9
Aspen Plus model (%)	14.97	20.50	13.30	50.32
Relative Error (%)	3.4	−7.3	−16.7	4.9

3. Results and Discussion

In order to analyze the effect of the co-gasification process parameters on syngas composition after adding a water–gas shift reactor, a sensitivity analysis to the steam-to-feed ratio (SFR) was performed. Moreover, the co-gasification process using air and steam as the gasifying agents is compared before and after the WGSR. The co-gasification parameters are fixed to maximize the hydrogen amount in the syngas according to the findings of Bourguig et al. [16]. Therefore, the gasification temperature is fixed at 750 °C and the feedstock mixture comprises 70% MSW and 30% biomass, as shown in Table 1.

3.1. Effect of the SFR

The steam-to-feedstock ratio plays an important role in the gasification process, mainly because it enhances the hydrogen content of the syngas [15,16]. Figure 3 shows the results obtained after adding different amounts of steam to the gasifier in order to reveal the effect of steam concentration on syngas composition.

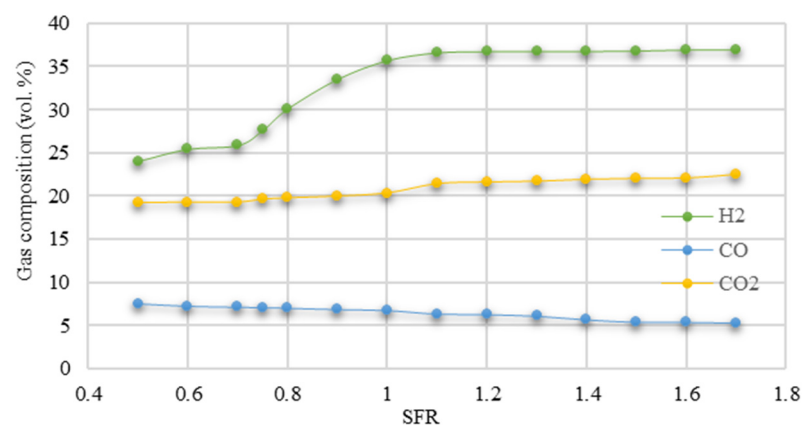


Figure 3. Effect of SFR on syngas composition.

Figure 3 shows that H₂ and CO₂ volume fractions increase, while the CO volume fraction decreases with increasing SFR. According to the Le Chatelier principle [41], the WGS reaction is favored in the forward direction as the concentration of the reactants (i.e., steam) increases. The results showed that the effect of the SFR on hydrogen concentration is relevant. Additionally, it is observed that the WGS reactor enhances the hydrogen content in the syngas and reduces the amount of carbon monoxide. These results are corroborated by the results of Pala et al. [19].

3.2. Effect of the WGSR

Temperature plays an essential role in the WGS reaction. This reaction is moderately exothermic and its equilibrium constant decreases with increasing temperature. The reaction is favored thermodynamically at lower temperatures and kinetically at elevated temperatures [30,31]. Therefore, the temperature range that maximizes the hydrogen yield in the WGSR should be defined. This is performed considering a co-gasification process using air as the gasifying agent. Figure 4 shows the effect of WGSR temperature on syngas composition, varying the temperature between 200 °C and 700 °C.

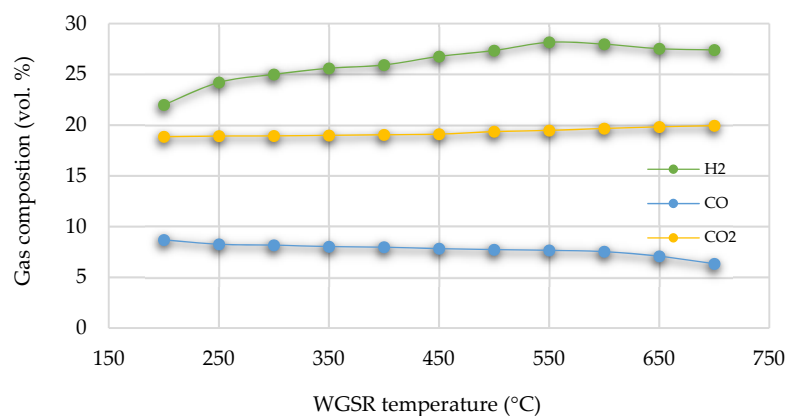


Figure 4. Effect of WGSR temperature on syngas composition.

Figure 4 shows that the volumetric percentage of H₂ and CO₂ slightly increases with the increasing temperature of the WGSR, while CO slightly decreases. The temperature range that maximizes hydrogen yield is 500 °C–600 °C. These tendencies agree with those reported by Pala et al. [19]. Figure 5 shows the difference in the amount of syngas produced through co-gasification before and after the WGSR.

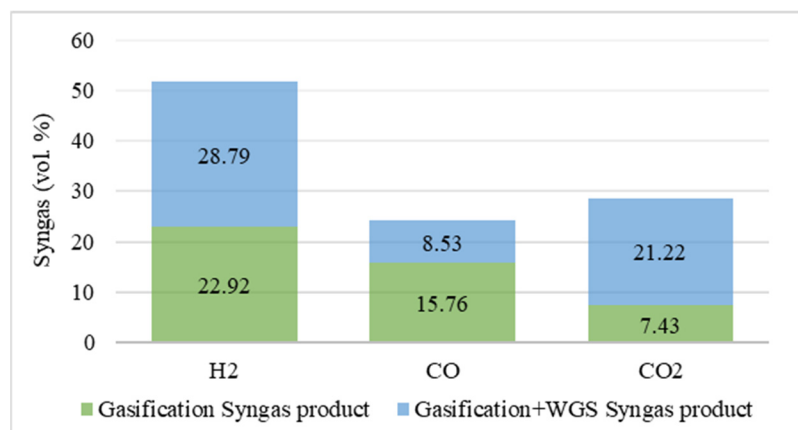


Figure 5. Syngas composition before and after the WGSR using air as the gasifying agent.

As shown in Figure 5, it is observed that the water–gas shift reactor enhances hydrogen and carbon dioxide while reducing the amount of carbon monoxide. These results agree with the earlier published data of Novais et al. [21]. Figure 6 shows the volume fraction of the main syngas species obtained through the steam co-gasification before and after the water–gas shift reactor.

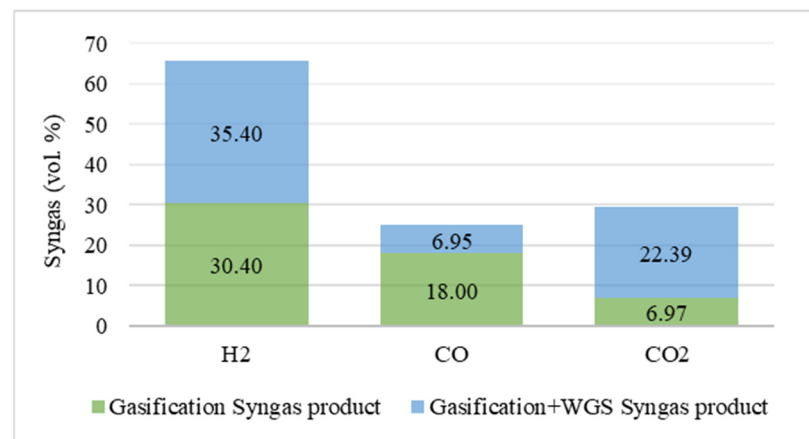


Figure 6. Syngas composition before and after the WGSR using steam as the gasifying agent.

Figure 6 shows that the water–gas shift reaction enhances hydrogen and carbon dioxide while reducing the amount of carbon monoxide in the syngas. It also demonstrates that the CO is mainly converted to CO₂ rather than H₂. Nevertheless, it can be concluded that the inclusion of a WGSR has an important impact on the quality of the syngas because it increases hydrogen and reduces carbon monoxide in the final composition of the syngas. These results agree with the earlier published data of Pala et al. [19].

4. Conclusions

In the current study, a comprehensive model is developed in Aspen Plus® that couples a co-gasification process of MSW and biomass to a water–gas shift reactor without steam injection. The main purpose of this work is to understand the advantages of using a WGSR to improve syngas quality. The results show an increase in hydrogen volumetric percentage for higher steam-to-feedstock ratios in the gasifier. The main findings of this study clearly indicate that the inclusion of a WGSR without steam addition increases hydrogen and carbon dioxide while reducing the amount of carbon monoxide in the syngas when using both air and steam as gasifying agents. This method is particularly relevant for high-moisture feedstocks since the WGSR can use steam produced in the gasification process. Moreover, the process may be further intensified by providing additional steam to the WGSR. This would raise the steam concentration in the reactants, favoring the forward WGS reaction, hence increasing CO consumption and H₂ generation. However, an economic assessment should be conducted on the final cost of the hydrogen due to the increase in operational expenditures caused by the addition of a steam generator.

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